

B.1 Densities & CDFs

L1

Continuous Distributions: Overview

$N \sim \text{discrete } \{1, \dots, 100\}$

$X \sim \text{continuous } (0, 100)$

$$P(N=n) = \frac{1}{100}, \quad n = 1, \dots, 100$$

$$P(N \leq n) = \frac{n}{100}, \quad n = 1, \dots, 100$$

$$P(X=x) = 0 \quad \forall x$$

$$P(X \leq x) = \frac{x}{100} \quad 0 \leq x \leq 100$$

The CDF, $F(x) = P(X \leq x)$ is valid for continuous distributions as well.

- For discrete,

$$E[X] = \sum x \cdot P(X=x)$$

but this term isn't valid for continuous, so -

for continuous,

$$E[X] = \int x \cdot f(x) dx$$

L2

Densities & CDFs

- The CDF of X is given by,

$$F(x) = P(X \leq x)$$

↳ this applies to all RV, whether they are discrete, continuous or mixed distributions.

- If X is purely continuous, then the density f_x is given by,

$$f(x) = \frac{d}{dx} F(x)$$

vice versa,

$$F(x) = \int_{-\infty}^{\infty} f(y) \cdot dy$$

- The properties of CDF,

$$1. \quad 0 \leq F(x) \leq 1$$

$$2. \quad \text{If } x \leq y \quad \text{then } F(x) \leq F(y)$$

$$3. \quad \lim_{x \rightarrow \infty} F(x) = 1$$

$$4. \quad \lim_{x \rightarrow -\infty} F(x) = 0$$

$$5. \quad \text{Since, } F(x) = P(X \leq x)$$

$$P(X > x) = 1 - F(x)$$

this is known as the survival function.

- Properties of densities -

1. $f(x) \geq 0$

2. There need not be any upper bound for $f(x)$.

3. $\int_{-\infty}^{\infty} f(x) dx = 1$

4. $\int_a^b f(x) dx = P(a < X \leq b) = F(b) - F(a)$

- x is a k^{th} percentile of X if $F(x) = k\%$. The median is the 50^{th} percentile, so $F(x) = 0.5$ is at the median.

Eg. CDF of X is
$$F(x) = \begin{cases} 0 & x < 1 \\ (x-1) - \frac{1}{4}(x-1)^2 & 1 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

$$P(X \leq 2) = F(2) = (2-1) - \frac{1}{4}(2-1)^2$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

$$P(1.5 < X \leq 2) = F(2) - F(1.5)$$

$$F(1.5) = 0.5 - \frac{1}{4}(0.5)^2 = 0.5 - \frac{0.25}{4}$$

$$= \frac{2 - 0.25}{4} = \frac{1.75}{4}$$

$$F(2) - F(1.5) = \frac{3}{4} - \frac{1.75}{4} = \frac{1.25}{4}$$

$$f(x) = F'(x) = \begin{cases} 0 & x < 1 \\ 1 - \frac{1}{2}(x-1) & 1 < x < 3 \\ 0 & 3 < x \end{cases}$$

L3. Mixed distributions.

- Suppose an insurance policy has a deductible of d and a payment limit of u . A customer (insured) has a loss of L .

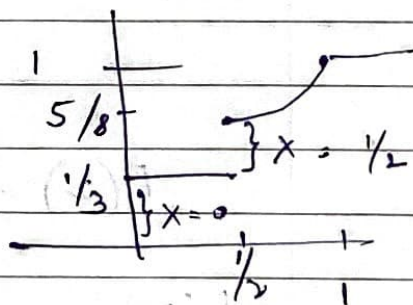
- Insured is responsible for d loss.
- Insurer has to pay that exceeds d up till the payment of u .
- After that (u), insured is responsible.

$$\begin{aligned} L < d & \quad \text{Pay} = 0 \\ d \leq L \leq d+u & \quad \text{Pay} = L-d \\ d+u \leq L & \quad \text{Pay} = u \end{aligned}$$

Here, 'pay' means payment made by the insurer. The premium refers to payment made by insured to insurer.

- Remember, sometimes $F(x)$ is not purely continuous $\Rightarrow$ there are possible jumps.

$$F(x) = \begin{cases} 0 & x < 0 \\ 1/3 & 0 \leq x < 1/2 \\ \frac{x^2+1}{2} & 1/2 \leq x < 1 \\ 1 & 1 \leq x \end{cases}$$



$$F(0) = \frac{1}{3}, \quad F(1/2) = \frac{5}{8}$$